GeMR : Multi-modal Interactive 3D Scene Composition In XR

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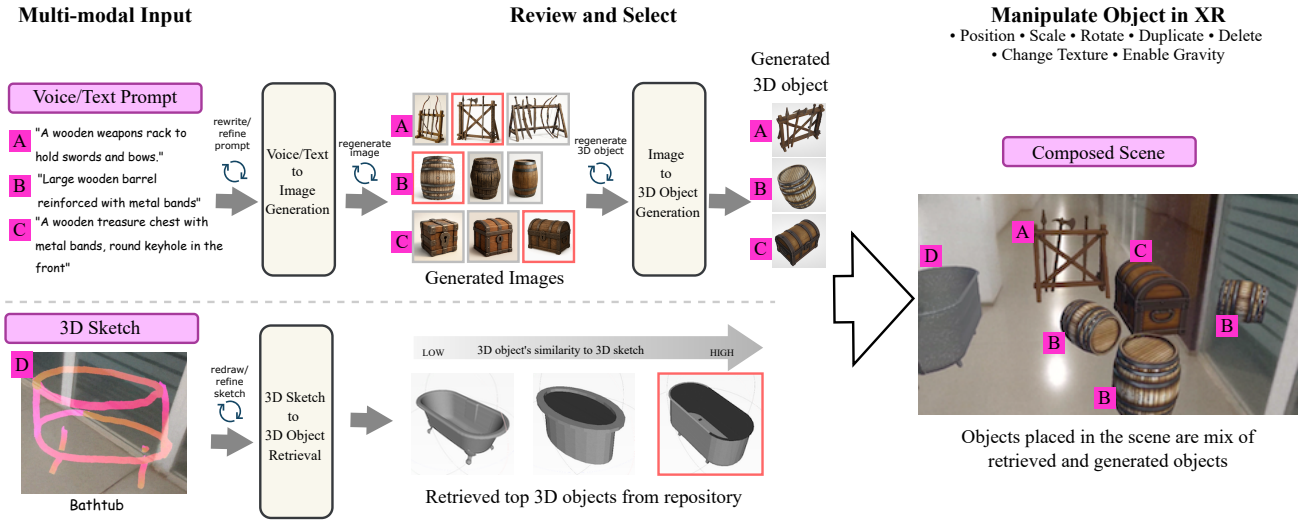


Figure 1: GeMR ('Ge'nerative 'M'ulti-modal X'R') is an interactive 3D authoring platform that empowers users to create novel scenarios and assets by translating intuitive multi-modal inputs such as audio, text, sketches, or gestures, into immersive 3D content. It enable users to retrieve assets with specific shapes or geometries from a library of pre-generated or artist-created 3D assets by using free-hand 3D VR sketches as search queries. Additional features include interactive composition, texturing, and physics integration, offering a streamlined, user-driven workflow for designing complex 3D environments.

Abstract

This paper introduces a novel mixed-reality (MR) 3D authoring platform designed to enable real-time, interactive composition of 3D scenes in augmented and mixed reality environments. The platform proposed and integrates two key innovations: a three-tier 3D asset generation workflow and a 3D sketch-to-shape retrieval algorithm. The three-tier architecture addresses the challenge of time-consuming 3D asset creation by allowing users to engage with the generation process at intermediate stages, enabling rapid validation and refinement of user inputs. This approach not only enhances interactivity but also accelerates the realization of user imagined assets. Complementing this is our 3D sketch-to-shape retrieval system, which leverages an intuitive, robust sketching module to query a library of generated or artist-created 3D assets. This mechanism facilitates precise retrieval of objects matching both the category and rough shape of a user drawn 3D sketch of the assets. The platform supports multi-modal, intuitive user inputs such as speech, text, and freehand 3D sketches, making it accessible to both novice and expert users for creating, retrieving, modifying, and composing 3D assets and textures. By combining generative models with intuitive interaction, the platform serves as a flexible testbed for 3D asset development and a tool to democratize 3D scene authoring. We show effectiveness of GeMR on various use-cases, including scenario composition for human-robot collaboration. We conducted a user study to evaluate whether our system effectively democratizes the 3D scene composition. We compared the performance of 3D modeling experts and novice users on a standardized scene creation task. Both novice and expert users reported equally good usability ratings and low perceived workload. Our findings highlight that an integration of generative models and multimodal

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intuitive inputs represents a significant leap in making the 3D scene authoring more accessible.

CCS Concepts

• **Human-centered computing** → *User interface design*; **Mixed / augmented reality**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Mixed reality, Generative AI, Scene composition, AIGC

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1 Introduction

Recent advancements in the Mixed Reality (MR) technology have significantly reduced hardware costs, paving the way for widespread adoption across various industries, including design, education, and entertainment. This has increased the demand for 3D content. Traditional 3D modeling relies on tools like Blender[5], which require expert-level skills, and the development cycle is slow and expensive [33] . Meanwhile, 3D modeling in MR has become popular due to its immersion, and spatial interface. But the current 3D modeling tools in MR and desktop systems rely on detailed sketching and sculpting skills to achieve the desired output [10]. These constraints limit the widespread participation in 3D content creation.

With the recent boom in generative AI (GenAI) space, 3D content generation through natural language prompts has become possible. Text-to-3D and image-to-3D generation techniques have opened up new possibilities for expert designers and novice users alike [37] [45]. GenAI, when combined with MR systems, can empower architects, educators, hobbyists, etc., to build rich immersive 3D scenes without the need of design skills training. Such developments open up new avenues for game development, interior design, floor planning, and many more. However, the existing text/image-to-3D asset generation pipeline are time consuming and are not suitable for interactive applications. Mixed Reality offers a unique advantage over 2D screens by enabling users to manipulate and perceive 3D content in a more intuitive manner with a sense of scale and depth [29]. Recently, there has been a surge in GenAI-based 3D content creation tools. However, most of the current tools remain constrained. They either demand highly specific text prompts or rely on content retrieval from a preloaded library of 3D assets, offering a limited scope for in-situ and spontaneous content generation capabilities. Many of these systems lack an interaction-first design, not embracing intuitive modalities such as hand gestures and freeform 3D sketching. Many current systems depend on text prompts alone. While text offers flexibility, it can be imprecise for describing 3D shapes and spatial arrangements. Furthermore, long generation times in generative AI models can disrupt the immersive flow of MR experiences. However, multi-modal inputs such as 3D sketching allow users to convey shape, structure, and context more intuitively, helping to bridge the gap between imagination and output [11].

We argue that to truly democratize 3D content creation in MR, we need a workflow that allow users to provide inputs in intuitive ways and elevating the need of being an expert in 3D designing. In addition to that the process must be interactive, and amenable to user feed backs at various intermediate stages. Such platform not only enhance the user experience but also ensure modularity, scalability, and extensibility. In this work, we present multi-modal interactive 3D scene composition platform in XR (dubbed as **GeMR** : **Ge**nerative **M**ulti-modal X**R**), that enables intuitive creation, placement, and manipulation of 3D objects in MR environments.

Our key contributions are as follows:

- A generative AI powered 3D authoring platform for interactive 3D scenario creation from user imagination.
- An multi-model intuitive interface to generate, retrieve 3D objects by voice, text and 3D sketch-based prompts on-the-fly. Offering the ability to locate, rotate and scale objects by hand. Along with the ability to manipulate the texture and physics.
- An enhanced 3D sketch-based retrieval algorithm for retrieving semantically similar 3D objects from a dataset.
- A modular and extensible software framework for generative 3D scene composition in MR.
- A user study to evaluate the system's usability, performance, and user satisfaction.

2 Related Works

In this section, we review multi-model scene composition tools, generative AI methods for 3D content generation, and sketch to shape retrieval methods.

2.1 Multi-modal Scene Composition Tools

Existing systems for AI-based scene composition either provide a desktop interface for scene building or do not have real-time 3D object generation in XR. Recent generative approaches such as Holodeck [40], demonstrate text-to-3D scene generation pipeline for embodied AI, but are restricted to desktop interfaces, limiting the usability for immersive and interactive applications. LLMER [8] demonstrates a speech-based in-place content generator system in XR but the generated objects are not realistic because they are composed of a predefined set of 3D shapes. GenDeck [35] presents a VR application pipeline for text-to-image-to-3D mesh generation but lacks real-time mesh generation. VRCopilot [43] observed that while describing an immersive scene, users do not have a concrete idea in mind. Showing intermediary low-fidelity representations such as images helped the users in refining the textual prompts. MS2Mesh [33] used a feedback system which uses image generation to help users refine their input prompts for 3D object composition in MR. Moreover, text-to-3D object generation is a computationally expensive task due to which it takes significantly more time than text-to-image generation. Such a long waiting time in VR may break the immersion and eventually draw the user away from the experience. Relying on a preloaded library of 3D assets for content creation in the scene limits the scope for in-situ and spontaneous content generation capabilities. LLMER [8] adopted an approach in

which if the asset is not available in the repository, a primitive 3D shape like a cube or a sphere or a cone is spawned in the environment to represent the missing object. This highlights the need of an on-the-fly object generation system. Similarly, MS2Mesh [33], VRCopilot [43], GenDeck [35] also depend on a predefined set of 3D objects available to the system, but did not discuss any strategies to handle cases when the desired asset is not available.

Most modern large-scale applications are made up of multiple services, also known as service-oriented architecture (SoA) [1]. A similar approach when followed to build an immersive scene composition tool would make it scalable to multiple users, extensible to rapidly add new features and fault-tolerant. Few methods rely on third-party services. [33] depends on the Meta Voice SDK [23] for text-to-speech transcriptions. [8] depends on OpenAI's cloud platform [24] for using LLMs. Such systems, when scaled up, may pose challenges like high tariffs on third-party AI service, binding with specific VR platforms, or privacy concerns because of the data outflow to third party servers. A review paper [27] mentions that the absence of a well-defined architecture is a significant barrier to adoption of existing tools. Moreover, [27] suggests cloud-edge deployment approach to solve latency and scalability issues with GenAI models for on-the-fly 3D content generation.

2.2 Generative AI for 3D Content Generation

Recent methods[30, 37, 45] have proposed several approaches for generating 3D objects from images, text, or 2D sketches. While these methods produce high-quality outputs, they do not support 3D sketching, which limits their applicability in immersive environments. For conveying user intent in virtual reality (VR) and accurately capturing spatial imagination, 3D sketching offers a more intuitive and expressive medium.

Recent methods[30, 37, 45] have shown progress towards generating 3D objects from images, text, or 2D sketches. Yang et al. [40] proposed a text-driven scene generation method that leverages the capabilities of GPT-4 [31] to produce an initial empty floor plan. This floor plan is then extruded into a 3D structure, and objects are populated into the scene by retrieving them based on textual descriptions. Chen et al. [9] introduced a technique for generating empty scene layouts using a generative adversarial network (GAN). Their system allows users to customize room textures and generates an empty floor plan which is then extruded to form a basic 3D room. Zinin et al. [16] presented a method that generates Blender scripts for scene creation. Their approach composes scenes by retrieving and placing objects from a predefined object library based on scene specifications. Several methods like TEXTure [28], Text2Tex [7], and InTex [32] have focused on generating textures for 3D assets. While effective, these techniques often struggle with maintaining multi-view consistency, leading to seams and artifacts. To address this, SyncMVD [19] proposes synchronizing the diffusion process across views to enforce consistency. Another significant challenge is that textures generated from 2D image models often have lighting and shading effects "baked-in". To tackle this, Paint3D [42] introduces a refinement stage to produce lighting-less textures, while DreamMat [44] generates full PBR material maps by using a light-aware diffusion model. Alternatively, TEXGen [41] trains a diffusion model directly on UV texture maps. More recently, methods like Make-A-Texture [39] have focused on dramatically accelerating the generation pipeline to enable interactive rates. We have adapted InTex [32] in out MR Authoring tool as it has decent quality texture generation power with highest inference speed across all openly available texture generation methods.

Across these methods, while substantial progress has been made in generating 3D objects and textures from sketches or other multi-modal inputs, there remains a notable gap in addressing the design and implementation of, interactive 3D workflow with user feedback, sketching interfaces and retrieval systems that facilitate effective user interaction.

2.3 Sketch-based Retrieval of 3D Content

Early methods, such as those in [12, 34], tackled shape retrieval using 2D sketches. While these approaches showed some success, they were limited by the projection of 3D geometry into 2D, causing ambiguity and lower accuracy, particularly for objects with complex spatial structures. The rise of immersive technologies has made 3D sketching in virtual environments a promising alternative. Giunchi et al.[13] and its follow-up [2] demonstrated the potential of mid-air 3D sketches for real-time retrieval, offering intuitive solutions for spatial design tasks. However, these systems used basic shape descriptors, leading to suboptimal retrieval performance. Luo et al. [21] introduced the first large-scale 3D VR sketch dataset paired with shape retrieval models. Their work compared representations like voxel grids, multi-view projections, and point clouds, finding that PointNet++ [25] embeddings for 3D sketch point clouds performed best. Nonetheless, this method relied on fixed-size point representations, which struggled with the variability of sketch inputs. To improve retrieval accuracy, Luo et al. [20] developed a structure-aware triplet loss that adjusts similarity margins based on shape fitting errors, enhancing fine-grained retrieval but still using fixed-point cloud representations. While these methods demonstrate the ability to generate and retrieve 3D objects from 3D sketch inputs, there is limited attention given to user-driven sketch-based scene composition or interactive retrieval systems that support intuitive scene creation.

3 GeMR : Our Proposed Platform

Our proposed platform (**GeMR**), consists of following four major components: (a) Multi-modal user interface for intuitive inputs and composition, (b) three-tier 3D asset generation workflow with quick validation and refinement, (c) robust 3D sketching in XR and sketch to shape retrieval, and (d) interactive 3D asset texturing. The platform is a Unity application running on the headset and a set of services running on different servers. Figure 2 contains a detailed system overview. Our platform enables on-the-fly scene composition in a mixed-reality setup. Users can create 3D objects and texture then from voice or textual prompts. Users can draw 3D sketches to retrieve objects. Each feature is hosted as a service as shown in the 2. All the services use custom API based on Python-Flask[1]. Our service-oriented approach makes the system highly flexible as we can change the AI models behind the features without making changes to the platform itself.

[1] https://flask.palletsprojects.com/en/stable/

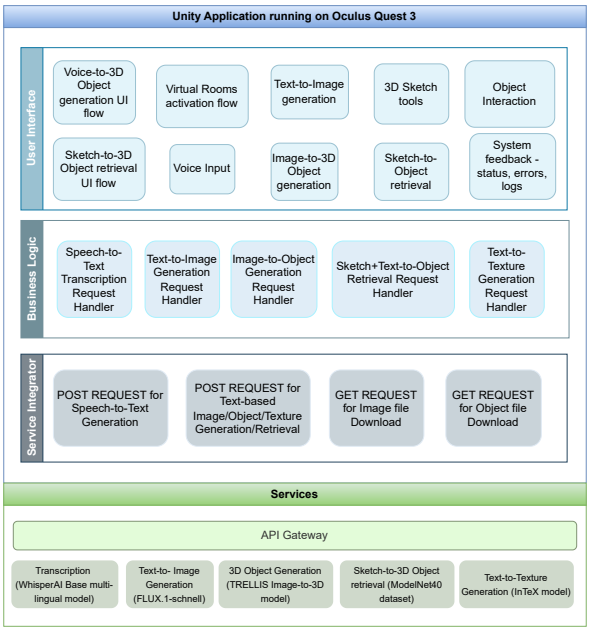


Figure 2: The system consists of two layers - *(A) Application Layer* – a Unity application running on Oculus Quest 3 – and *(B) Services Layer* – a set of services, each running independent of the other. The layer A follows an object-oriented architecture. Each components are conceptually classified under three sections as - *(A1) User Interface, (A2) Business Logic and (A3) Service Integrator*. Layer B, consists of transcription service, generative AI services for image generation, 3D object generation and retrieval and texture generation. The API gateway server is implemented in Flask. It reroutes incoming requests to the suitable service. Section A1 contains all the visual components and methods for enabling user interactions. Section A2 serialize and deserialize the requested data based on the user interactions. Section A3 is responsible for creating requests to the API Gateway.

In the following section we first discuss the interface for multi-modal inputs, then three-tier 3D asset generation and texturing, followed by sketch to shape retrieval.

3.1 Multi-Modal User Inputs

The user interface of the proposed platform enables users to write text prompts via voice input, or draw a 3D sketch of the desired object, or manipulate the spawned objects. There are three input options available in the interface – *voice and text input* for object generation, *3D sketch input* for object retrieval, and *voice/text input for texture generation*. Each input option has an interaction flow (as shown in the figure 4) which can be classified into three steps – *input step, review step, and object transformation and manipulation step*. To generate a *3D object from voice or text prompt*, the user clicks on the mic button on the panel. The user speaks in the description. If successful, the output will be visible in a text box, which the user can edit. Then submit the text by clicking on the generate button. When the user submits a prompt to generate a 3D object, the interface first shows an image of the desired 3D object. The user will confirm whether the generated image resembles the desired 3D object or whether a new image needs to be generated before initiating the 3D generation step.

Similarly, to *retrieve a 3D object by drawing a 3D sketch*, the user first enables the sketching mode. A movable sketch orientation gizmo appears to assist the user in defining the orientation of the sketch they intend to draw. After the user freezes the gizmo orientation, user can draw a sketch through pinch and poke gestures (discussed later in this section). The sketch toolbar appears in the environment which provides undo, redo and deletion options. After completing the sketch, the user confirms the sketch category and hits the retrieve button. The top three matching objects are retrieved from the repository for the user to review and select one. On selection, the selected object spawns in environment for the user to manipulate and transform.

After a 3D object is spawned in the environment, the user can change the object's properties such as *enable/disable gravity*, *generate and apply a texture*, *duplicate* or *delete* the object. These options are listed in a menu which the user can call by hovering a palm over the object. Moreover the user can transform the position, rotation, and scale of the generated 3D object intuitively by hand without needing joysticks/VR controllers. Generating a new texture is similar to voice/text input flow for object generation. After speaking/typing in the texture description, an image of the generated texture is shown. If the user is satisfied with the output, they can press on the image to apply the texture, else they can restart the process.

There are two types of gestures - pinch and poke (refer to Fig.3). Here is a list of different functionalities described along with the gestures required:

Move, rotate and scale a 3D mesh : To select and transform a mesh's position and rotation, a user should move their hand(s) onto the mesh and use a pinch gesture to grab it. Then while maintaining the gesture, a user can rotate or move the mesh. To scale up the mesh, the user can select and stretch. To downscale the mesh, the user should move his hands in the opposite direction.

Draw a sketch and retrieve mesh : Sketching is divided into two parts; a pinch gesture on the left-hand hand is required to enable the sketching mode, and the poke gesture on the right-hand hand is used as a stylus. There is a sketching menu that provides sketch editing options. To retrieve a mesh, the user has to choose an object category from a drop-down (these categories act as a context for the sketch for faster and more accurate mesh retrieval). Then submit the sketch and the category together.

Manipulate mesh attributes : When a user hovers their hand close to a mesh in the scene, a menu pops up, which provides options to tweak specific attributes of that object. These attributes include texture, enable or disable gravity, duplicate and delete, and resetting the object's position. To generate a new texture, the user can type in the description of the texture.

3.2 Three-tier workflow for 3d asset generation

Direct text-to-3D generation is a time consuming task, additionally it is difficult for a user to provide all fine geometric and appearance

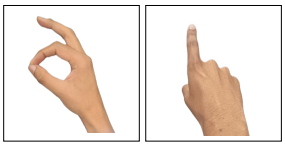


Figure 3: (a) "Pinch" gesture is used for grabbing and transforming an object. It is also used to enable/disable sketching when in sketch mode; (b) "Poke" gesture is used for clicking on UI buttons. It is also used as a stylus when in sketching mode.

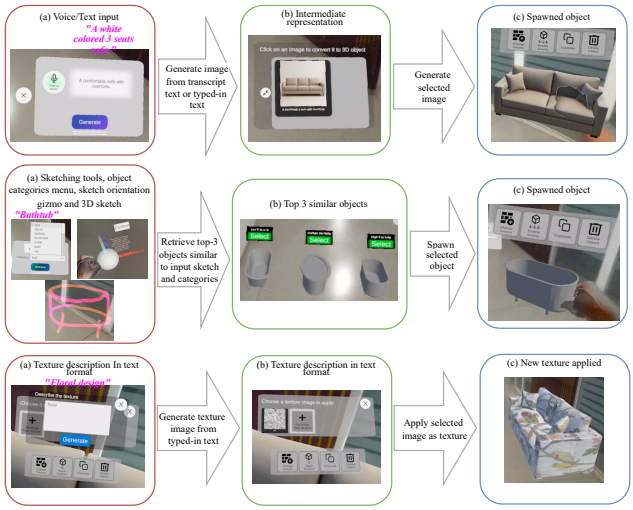


Figure 4: From top to bottom -> (1) UI flow for *voice input*; (2) UI flow for *sketch input*; (3) *UI flow for texture generation.* From left to right -> RED: *Input Step*; GREEN: *Review Step*; BLUE: *Object transformation and manipulation step.*

details in the text. To elevate this issue we use image as an intermediate representation to validate and refine input or asset description before generating 3D asset. Specifically, we propose, three-tier 3d asset generation workflow, allowing user feedback at intermediate stages for quick validation and refinement. In the first tier, user provide 3D asset description in the form of audio(voice) that is translated to text, at this stage user can refine the transcription or re-record the voice description. Once satisfied, text is used to generate 2D image using text-to-image generative models. User can validate if the generated image is matching the user's imagination, if not, user can regenerate the image. If satisfied, in the third tier , the image is used to generated 3D asset. In our experiments, we found the total turn around time to generated the desired imagined 3D asset using our proposed three-tier approach is less, compared to direct text-to-3D asset generation. This is primarily, in our proposed workflow the number of calls for 3D asset generation is on average 1, while in direct text-to-3D generation is much higher than 1. We discuss, speech-to-text, text-to-image and image-to-3D generation modules in the following sections.

Speech-to-Text Module. To enable hands-free, natural interaction in immersive environments, our system integrates a speech-to-text (STT) module powered by OpenAI's Whisper Base multilingual model [26]. Whisper is chosen for its strong robustness to accents, background noise, and reverberation—common in XR environments—making it well-suited for real-time use. The transcription service runs as a dedicated Python Flask micro-service on the network, allowing the XR application to offload audio processing and maintain interactive performance. This modularity ensures flexibility in system design, enabling easy upgrades or scalability without altering the XR interface. The user experience is intuitive: the user taps a microphone button within the scene to activate voice input, speaks a command or object description, and the transcribed text is immediately displayed on-screen. Users can verify, edit, or confirm the text before it is submitted for object generation, texture updates, or system-level commands. This feedback loop ensures reliability while preserving the flow of interaction. By eliminating the need for controller-based typing or complex menu navigation, the system reduces friction for novice users and supports fast, expressive scene creation. This speech input module forms a key part of our multi-modal interaction framework. It complements other input modes such as sketching and manual text entry, giving users the freedom to switch modalities depending on context. The inclusion of real-time speech enhances accessibility, enabling users to focus on creative exploration without leaving the immersive environment. As such, Whisper-driven transcription not only improves system usability but also aligns with our goal of democratizing 3D scene authoring in XR.

Text-to-Image and Image-to-3D Generation Module. Our platform employs a two-step approach for 3D asset generation, a deliberate design choice that prioritizes user intent and computational efficiency within an interactive XR environment. Direct text-to-3D models, while fast, operate as a "*black box*" which can lead to misinterpretations of the user's prompt and result in wasted computational cycles on undesirable assets. This trial-and-error process is ill-suited for the fluid, real-time feedback loop required in an immersive authoring tool, where long pauses can break creative flow and cause user frustration. The first step addresses this by using a rapid text-to-image model, FLUX.1-schnell [17], to generate one or more 2D images from the user's prompt. This model is optimized for speed, providing a near-instant visual preview that serves as a critical verification checkpoint. By presenting these images, the system allows the user to confirm that the visual representation matches their creative intent before committing to the more resource-intensive 3D generation step. This *user-in-the-loop* approach is fundamental to our system, ensuring that computational resources are used effectively and empowering the user with greater control over the final outcome. Once the user selects a satisfactory image, the second step begins, where a state-of-the-art image-to-3D model, TRELLIS [38], *lifts* the 2D images into a high-quality 3D mesh. This two-step process is not a compromise but rather the prescribed best practice for achieving the highest quality results. The developers of TRELLIS explicitly recommend this workflow, noting that their image-conditioned models produce more creative and detailed assets than their text-conditioned counterparts due to the limitations of 3D training data. Our architecture therefore creates

a symbiotic relationship: FLUX.1-schnell [17] rapidly generates a high-quality visual prompt that plays to the strengths of TRELLIS, maximizing the fidelity of the final 3D asset while keeping the user in control.

3.3 Sketch-based 3D Object Retrieval Module

Traditional 3D modeling tools such as Blender[5], Maya[3], and ZBrush[22] etc require substantial prior knowledge, making them unsuitable for intuitive 3D sketching. However, 3D sketching offers a more natural and expressive way for users to convey their imagination—especially as it captures topology in a view-independent manner, unlike 2D sketches. While recent advancements in generative AI enable 3D model generation from text, textual descriptions alone are often insufficient to fully capture a user's intent. Sketching, a historically effective means of expression, remains vital in this context. To address these challenges, we propose an immersive VR-based 3D sketching interface that not only allows users to intuitively draw in 3D but also retrieves semantically and structurally similar 3D models from a predefined dataset. Our retrieval system builds upon the work of Luo et al.[21], introducing key improvements. Luo et al. [21] could only process 80% of ModelNet10 which has 10 categories of object. We extend their dataset to include 10 more categories from ModelNet40[36]. Furthermore, while Luo et al.[21] used L2 distance for retrieval, we replace it with cosine similarity, which better captures the angular closeness of feature embeddings and proves to be more effective for semantic similarity. As shown in Table 1, this modification leads to significant improvements in top-1,2,3 retrieval accuracy.

Method	**Top1**(↑)	**Top2**(↑)	**Top3**(↑)
Luo et al.	32.3	40.1	48.4
Proposed Method	35.3	43.4	50.1

Table 1: Comparative analysis between Luo et al[21] and Proposed Method with respect to Top1, Top2, and Top3

3.4 Text-to-Texture Generation Module

A dedicated, on-the-fly text-to-texture module is essential for a flexible and interactive creative workflow. Users may not be satisfied with the initial texture of a generated asset or may wish to apply new textures to existing meshes. Our system provides this capability, allowing users to apply, modify, and refine textures using simple text prompts. This transforms texturing from a static, one-time operation into a dynamic and iterative process, enabling rapid experimentation with different material styles without leaving the immersive environment. This procedural, non-destructive workflow is crucial for maintaining creative momentum and lowering the technical barrier for users. Many existing text-to-texture methods, which *paint* a mesh from multiple viewpoints, often struggle with maintaining global coherence, resulting in visible seams and inconsistent artifacts that degrade asset quality. While some approaches attempt to solve this, they can introduce architectural complexity or still suffer from other issues. These challenges highlight the need for a solution that balances speed, quality, and, most importantly, consistency. To meet these requirements, our system integrates InTex [32], a framework designed specifically for interactive text-to-texture synthesis. InTex addresses the consistency problem by using a unified depth-aware inpainting model trained on 3D data, which reduces artifacts and improves generation speed. This gives users granular control to refine textures on the fly, perfectly aligning with the real-time, user-driven design philosophy of our XR authoring platform.



Figure 5: living room composed using GeMR



Figure 6: game arena composed using GeMR

4 Experiments and Analysis

In this section, we illustrate the wide range of objects and scenes that one can compose using our proposed method **GeMR**. Our method allows user to generate and composition scene for any tasks in XR. Particularly, for proof-of-concept we show virtual room and game environment design, and virtual scenario composition for human-robot collaboration. We present an detailed user study to validate the effectiveness of our proposed method.

4.1 Virtual room and game environment design

A game or an interior designer can use *GeMR* to create a draft environment, and quickly iterate over different ideas by generating and manipulating the objects. For example, a living room composition (see figure 5). It consists of multiple chairs (generated by voice and then duplicated multiple times), a sofa (generated from voice, and a texture is applied it), a center table, a plant (both generated from voice) and a big cushioned chair (generated from sketch, and a texture is applied to it). Another example is a RPG game arena (see figure 6). It consists of voice generated medieval themed objects like wooden barrels (generated by voice and then duplicated multiple times), a swords (generated by voice) and bow stand (generated by voice), a wooden treasure chest (generated by voice), and bathtub (generated by voice and text generated metal-like texture applied to it).

4.2 Human-robot Collaboration (HRC) Training

The highly modular architecture of our work enables developers to integrate other services. We present a case where we connected our system with a robot planning module which consisted of a ROS (Robot Operating System) server and motion planner. This setup enabled in-place HRC testing, supporting both industrial and creative applications enabling in-place collaboration **(see fig 7)**. The system allows the users to build the scene and integrate the robot for collaborative testing. The robot's motion planner, guided by the virtual world state, robot joint data, and ROS inputs, generates actions for collaboration tasks, repeating in a feedback loop. User places markers to guide the robot to understand movable and non-movable objects. For example, the tables are non-movable while the 3D objects are movable. The user can further interactively

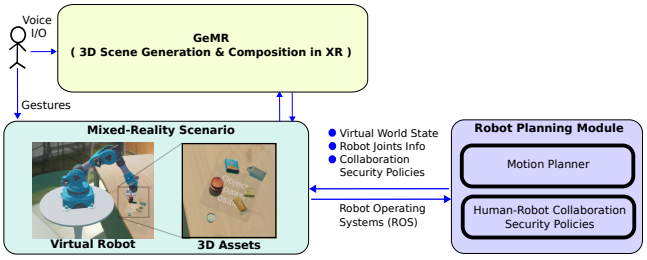


Figure 7: Interactive in 3D scenario generation and in-place human-robot collaboration in MR.

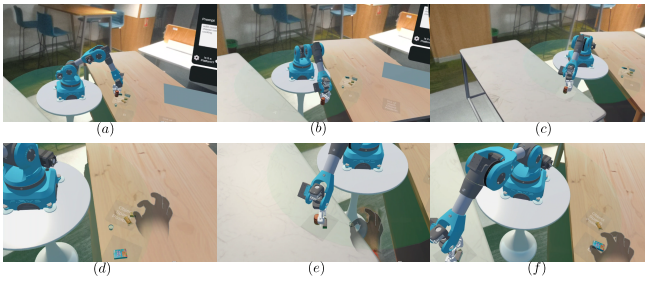


Figure 8: Human-Robot Collaborative Sorting Task: Human and robot are collaboratively picking and placing objects from place on one table to another.

place objects outside the reach of the robot. This is for the robot to understand, and respond that certain objects are outside reach, and request the user to pick and place on its behalf. The user also places a marker to specify the target location for each of the object category for sorting. Once the user finalize the composition of 3D objects and setup the sorting specification, the collaboration can be started by indicating robot to start operating by pressing a start button. (see fig. 8).

4.3 User Study

We conducted a user study to evaluate the effectiveness of **GeMR** in assisting users with immersive scene composition tasks. Participants answered a set of questionnaires and completed a scene composition task.

Apparatus: The study was conducted in a well-lit 5 m x 4 m room, providing ample space for scene composition. The device used was an Oculus Quest 3 headset.

Participants: A total of 10 participants (mean age: 27.9, 8M, 2F) were recruited. Five reported expertise in 3D modeling using Blender. All participants had prior experience with AI prompt writing and immersive VR applications.

Procedure: We first demonstrated **GeMR**'s features through a recorded video, followed by a 10-minute exploration period to help participants get comfortable. They then performed the scene composition task using **GeMR** in mixed reality mode (with passthrough on Oculus Quest 3). To ensure task uniformity, we provided a 2D layout of a target scene—a living room. Participants could use all modalities to compose the scene but followed the layout, creating only the specified objects. They decided descriptions, and could rotate or scale objects to build a meaningful setup. After completing the task, each participant filled the NASA-TLX and System Usability Scale questionnaires and participated in a semi-structured interview.

	Novice	Expert	Overall
Mental Demand	3.4(1.52)	2.80(0.84)	3.10(1.20)
Physical Demand	4.86(0.78)	5.66(1.04)	5.26(0.96)
Temporal Demand	1.88(0.70)	1.77(0.31)	1.81(0.51)
Performance	6.83(2.12)	7.14(1.01)	6.99(1.57)
Effort	3.43(0.66)	3.37(1.15)	3.40(0.89)
Frustration	2.43(0.69)	2.17(0.80)	2.30(0.72)
Score	3.80(0.40)	3.82(0.32)	3.81(0.34)

Table 2: NASA-TLX survey Mean(Standard Deviation) of scores for Novice and Expert participants, and combined. Scale 1-10.

Measurements To assess the user perceived effort, we used the NASA-Task Load Index [15]. Participants rated the items on a 10-point Likert scale (1=Strongly Disagree, 10=Strongly Agree). We computed the Raw-TLX scores as the unweighted average of the six dimensions as suggested in the [14]. To asses the usability of our system, we used the System Usability Scale [6]. Participants rated the items on a 5-points Likert scale (0=Strongly Disagree, 4=Strongly Agree). The final scores were calculated using the standard calculation procedure (which converts the final scores to the range of 0-100). We conducted a semi-structured interview after scene composition task completion to gather feedback regarding the overall experience, challenges faced, and the scope for improvement. The interview consisted of a 5-item open-ended questionnaire and it was conducted in a conversational manner. Below are the questions asked in the semi-structured interviews:

- Q1: Can you describe your overall experience using the 3D Scene Composition Suite for creating 3D content?
- Q2: Which features or modes of interaction (voice input, 3D sketching, hand gestures) did you find most helpful or intuitive?
- Q3: Did you encounter any difficulties or frustrations during the task? How did you manage or overcome them?
- Q4: How did your prior experience with 3D modeling or prompt writing affect your use of the system?
- Q5: What suggestions do you have for improving the system to better support different levels of user expertise?

4.3.1 ***Quantitative Results.*** **(a) Perceived Workload :** Participants reported low workload levels on average, with a mean NASA-TLX score of 3.81 (0.34) (scale 1-10). Mental and physical demands were moderate, temporal demand was relatively low, suggesting that the participants felt minimal time pressure. Mean and Standard Deviation for each sub-scale are reported in the table 2.

(b) Perceived Usability : Table 3 presents the mean SUS scores and standard deviation reported by the novice and expert groups. The overall mean SUS score was 73(Standard Deviation=6.54) which according to [4] is considered a good and acceptable score. Novice users averaged 73.5(Standard Deviation=4.87), while expert users averaged 72.5(8.48), suggesting that the system provided a uniformly positive user experience irrespective of prior 3D modeling expertise.

	Novice	Expert	Overall
Score	73.50(4.87)	72.50(8.48)	73(6.54)

Table 3: SUS survey Mean(Standard Deviation) of scores for Novice and Expert participants, and combined. Scale 0-100.

(c) Impact of 3D Modeling Expertise : Experts reported a slightly higher physical demand score (M=5.66) compared to novice participants (M=4.86), potentially due to experts' tendency to perform more precise manipulations. Novice participants reported slightly higher mental demand scores (M=3.4) then experts (M=2.8). Some novices' first ever experience with 3D modeling of any kind in the digital space might have contributed to mental load. Frustration, effort and overall scores varied slightly for both the groups, suggesting that multi-modal input strategies effectively minimized usability barriers even for novices. Performance ratings remained high between skill levels (Novices: M=6.8, Experts: M=7.2) suggesting that our system democratizes 3D content creation for both experts as well as the novices.

4.3.2 ***Qualitative Results.*** After analyzing the responses from the quantitative surveys, we conducted qualitative interviews to gain deeper insights.

(a) Overall Experience : Novice participants appreciated our **GeMR** for 3D scene composition. *P3* stated that "With this tool I can quickly prototype my ideas. Traditional software like Blender are very intimidating to me". *P4* and *P5* also had similar views. On the other hand, Expert participants appreciated the ease of use and ability to quickly build 3D environments with our tool, they pointed out the need for more customizable interface and more features for 3D content manipulation. *P7* mentioned "It is very helpful tool for quick prototyping because that is all we need sometimes. But it lacks Blender like features for manipulating a mesh". Another expert participant said, "It would be great if I could arrange the tool panels according to my comfort and store the settings."

(b) Intuitive Features : The novice and expert participants liked the free-form 3D sketching as well as the ability to transform 3D objects by hand. One expert user *P7* reported the 3D sketching physically demanding given that it requires full body movement to draw the sketch. *P7* stated "There should be a way to draw the sketch without full body movement. I am not used to that. In desktop applications, such movements are not required".

(c) Challenges and Frustrations : Voice input was acknowledged as beneficial but the transcription was occasionally inconsistent which caused frustration. A novice user *P1* said "Voice input saved the effort of typing but sometimes the transcription was not correct". *P7* said "I tried multiple times to clearly speak my words but still transcription was not working properly". Expert users further expressed the need for faster generation time. All the users expressed a desire for presence of tutorials and guidance while interacting with the environment.

(d) Role of Prior Experience Prior experience in 3D modeling did not impact the interaction strategies significantly. Both sets of participants reported that they equally relied on sketching as well as textual input. However, three participants (*P1, P4, P8*) reported difficulty in composing accurate prompts (voice/text input) because they occasionally used the generative AI systems like ChatGPT. This suggests that prior experience in interacting with the large language models may impact a user's performance. Expert participants expected finer control over the composed environment, just like what they experience in tools like Bender and Autodesk Maya.

(e) Suggestions For Improvements Novice users recommended enhancements like guided tutorials. Experts suggested customizable controls, faster AI generation times, better interface for 3D sketching, and expanded advanced editing features to support complex tasks. All users suggested improving the transcription accuracy.

5 Discussion

Outcomes of the quantitative and qualitative analysis underscore the effectiveness of **GeMR** in democratizing 3D scene composition in MR. Moreover, the results highlight that expert and novice users found the system to have high perceived usability and low perceived workload. In line with previous work [18], our results reflect how the interaction-first, multi-modal input design successfully mitigated the complexity associated with 3D modeling tools. Experts appreciated the intuitive design and rapid prototyping capabilities but desired deeper control and precision tools, mirroring traditional workflows. This explains the high variance in SUS scores within the expert group.

The moderate physical demand reported in the NASA-TLX scores can be attributed to full-body movements and hand gestures required for sketching and interacting with the environment. Relatively higher physical demand ratings by expert users reflect their tendency for more precise, fine-grained adjustments in 2D software in which full-body movements are not required. Still, frustration levels remained low across all participants, suggesting that the system's interaction design effectively mitigates potential barriers.

One expert participant (P6) reported a low SUS score, which could stem from their expectation of advanced precision controls typically found in professional 3D modeling tools. It indicates expert users' desire for an interface which lowers physical effort and gives more control over the 3D model, aligned with their experience with traditional tools like Blender. However, the overall SUS score of 73 (novice=73.5, expert=72.5) falls under a "Good" usability range based on standard benchmarks [4]. The negligible difference between novice and expert users' SUS ratings validates the goal of democratizing 3D content creation.

6 Conclusion

We developed **GeMR**, a modular, multi-modal authoring tool designed to democratize 3D scene composition in Mixed Reality by combining generative AI with intuitive interaction techniques. Through use cases, we demonstrated the versatility of the system in applications from virtual room design to collaborative robot training. These real-world scenarios underscore the practical utility of **GeMR** for creative and industrial contexts. Our user study validates the system's accessibility and effectiveness. Both novice and expert participants reported low cognitive workload and high usability, with minimal variation in performance and satisfaction across expertise levels. These results highlight the power of multi-modal input—text, speech, and 3D sketching—in lowering barriers to entry for 3D content creation. The system balances ease of use for beginners and sufficient expressiveness for professionals seeking

rapid prototyping capabilities. **GeMR** demonstrates how multi-modal interaction and generative AI can simplify and democratize 3D scene composition in mixed reality. By combining text, speech, and sketch inputs with intuitive hand gestures, the system enables rapid, accessible 3D content creation for both experts and novices. Its modular architecture ensures extensibility, while user studies confirm high usability and low workload. This work marks a step toward inclusive design tools that empower a broader spectrum of creators.

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